Muhammad H. Rahman

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TECHNICAL SKILLS

Programming languages: Python, JavaScript, C#, C++, Java, Visual Basic, SQL
Front-End technologies: React.js, Next.js, Vite.js, Bootstrap, HTML/CSS, Qt
Back-End technologies: ASP.NET Core, Django, FastAPI, Express.js, PostgreSQL, Microsoft SQL Server, MongoDB
Developer tools: Git, Azure DevOps, Docker, AWS, Render, Railway, Vercel, Visual Studio, Claude Code, Amazon Q

PROFESSIONAL EXPERIENCE

SQA Analyst I, Geographic Solutions, *Remote* — June 2024 – Present

- Built C# automation suite with xUnit and Selenium WebDriver covering registration and resume builder flows as part of the migration from legacy VB scripts to a modern test framework
- Refactored VB test scripts for Reports module during WID 3.0 implementation, cutting team smoke test time by 50%
- Refactored deprecated VB regression tests for the job search UI, recovering ~8 hours per sprint on regression cycles
- Wrote parameterized Pytest API suites for Data Services team, adding coverage to aggregate labor market endpoints
- Reproduced XSS and CSRF vulnerabilities with crafted HTML payloads, partnering with developers on remediation
- Pair-program in a full-stack .NET codebase, contributing code review feedback and implementation suggestions
- Authored comprehensive Core module test documentation used to onboard 3 new client teams, communicating directly with client stakeholders and providing demos

SOFTWARE DEVELOPMENT PROJECTS

Gallerium - https://github.com/muhammadxrahman/gallerium
Real-time, offline-first planetarium PWA that computes space object positions on-device via published astronomical algorithms

- Built a from-scratch astronomy engine in TypeScript computing apparent positions of ~9,000 stars, 7 planets, the Sun, and Moon via Meeus and Keplerian theory and a topocentric pipeline (precession, sidereal time, refraction)
- Engineered real-time satellite tracking with SGP4 propagation and topocentric look angles, showing each satellite only when sunlit and the observer's sky is dark
- Developed dual Canvas 2D renderers (top-down dome and gnomonic AR), offloaded the ~9,000-star per-second compute to a Web Worker, and shipped an offline-first PWA with 281 tests and CI/CD auto-deploy to GitHub Pages

Lux Server - https://github.com/muhammadxrahman/lux
OpenAI-compatible LLM inference server for Apple Silicon (MLX/Metal)

- Built a FastAPI server that runs Llama 3.2 locally on Mac GPUs and serves it over the OpenAI API, with live token streaming and Prometheus metrics for throughput and latency
- Added continuous batching so simultaneous requests share each GPU pass instead of waiting in line, raising throughput from 88 to 231 tokens/sec (1→8 concurrent users) while keeping streaming response time low
- Wrote a custom C++ sampling kernel (temperature/top-k/top-p) checked against a Python reference in CI; profiled it, found the first version was 3x slower than NumPy, and fixed the data handling for a 5.6x speedup

ThreaDolphin - https://github.com/muhammadxrahman/threadolphin
Network packet analyzer for debugging TCP/IP communications that captures, filters, and exports traffic in Wireshark format

- Built multi-threaded packet capture engine in C++ using libpcap/Npcap and Qt6, implementing circular buffer with configurable limits (1K-100K packets) to prevent memory overflow during sustained high-volume captures
- Implemented TCP stream reassembly using 4-tuple connection tracking to rebuild application-layer conversations from fragmented packets, with bidirectional payload view for protocol debugging
- Added real-time, non-blocking filtering (substring search across protocols, IPs, and ports) plus PCAP import/export and native macOS/Windows builds for Wireshark interoperability

Scribble - https://github.com/muhammadxrahman/scribble
Real-time collaborative document editor with AI writing assistance that enables teams to co-edit with live cursors

- Architected full-stack application using ASP.NET Core 8, Next.js 16, and PostgreSQL, deployed on Railway and Vercel with SignalR WebSockets for real-time collaboration and live cursor tracking across multiple concurrent users
- Built ProseMirror rich text editor with 17 formatting options supporting 50,000-character documents with auto-save, PDF export, and granular Read/Edit permissions via JWT authentication
- Developed rate-limited Python FastAPI microservice integrating OpenAI GPT-4o-mini for AI text generation with private network communication between services

EDUCATION

The University of Texas at Austin, MS in Computer Science — 2028
Florida Atlantic University, BS in Biological Sciences, BS in Neuroscience and Behavior — 2022